Tutorial 13: Redox Reaction and Electrochemistry

Background:

1. Redox reaction — or oxidation-reduction reaction is reaction that involve changes in the oxidation state of the reactants as a result of exchange of electrons between them.

Reductant: the compound donates electrons and will be oxidized;

Oxidant: the compound gets electrons and will be reduced;

Reduction

$$Ag^+ + Ni(s) \rightleftarrows Ag(s) + Ni^{2+}$$

Oxidation

Oxidation state(O.S):	+1	0	0	+2
Role:	Oxidant	Reductant	Reduzate	Oxydate
Reaction:	Reduced	Oxidized		

2. How to balance Redox Equations?

By half-cell method:

1). Write unbalanced equations for the two half-reactions (consider oxidation numbers);

2). Balance atoms other than H and O;

3). Add appropriate number of electrons;

4). Balance O with H_2O;

5). Balance H with H^+;

6). If an acidic solution, skip to step 7; if a basic solution, then add equal number of OH^- to both sides to cancel all the H^+;

7). Multiply balanced half-reactions by appropriate coefficients so that the number of electrons are equal;

8). Add the half-reactions together and cancel species that appear on both sides to get the balanced Net Equation.

For example,

$$M_nO_{2(s)} + Cl^-_{(aq)} \rightarrow Cl_{2(g)} + Mn^{2+}_{(aq)} \text{ Under acidic conditions}$$

1) Write down the two half-reactions, respectively

Oxidation half Reaction:

$$Cl^-_{(aq)} \rightarrow Cl_{2(g)} + e$$

Reduction half Reaction:

$$MnO_{2(s)} + e \rightarrow Mn^{2+}_{(aq)}$$

2) Balance atoms other than H and O

$$2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + e$$

3) Add appropriate number of electrons

$$2Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots (1)$$

$$MnO_{2(s)} + 2e \rightarrow Mn^{2+}_{(aq)}$$

4) Balance O with H_2O

$$MnO_{2(s)} + 2e \rightarrow Mn^{2+}_{(aq)} + 2H_2O_{(l)}$$

5) Balance H with H^+

$$4H^+_{(aq)} + MnO_{2(s)} + 2e \rightarrow Mn^{2+}_{(aq)} + 2H_2O_{(l)} \text{ (Under acidic conditions)} \cdots (2)$$

6-8) The electrons are balanced, add Equation (1) and Equation (2)

$$4H^+_{(aq)} + MnO_{2(s)} + 2Cl^-_{(aq)} \rightarrow Mn^{2+}_{(aq)} + Cl_{2(g)} + 2H_2O_{(l)}$$

Exercise: $N_2H_{4(aq)} + MnO^-_{4\,(aq)} \rightarrow MnO_{2(s)} + NO_{(g)}$ Under basic conditions

3. How to predict does a redox reaction take place?

$Cu + AgNO_3 \longrightarrow$?

$Cu + Zn(NO_3)_2 \longrightarrow$?

$Fe + Cu(NO_3)_2 \longrightarrow$?

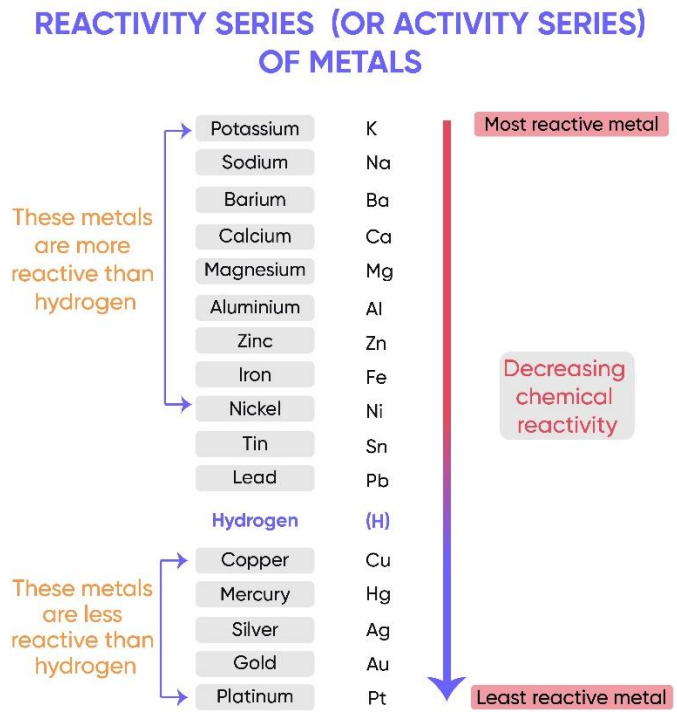
$Zn + Cu(NO_3)_2 \longrightarrow$?

$Al + Cu(NO_3)_2 \longrightarrow$?

$Mg + Cu(NO_3)_2 \longrightarrow$?

$Hg + Cu(NO_3)_2 \longrightarrow$?

Method I: Using the Activity series of metals



4. Standard Reduction Potentials (E^o)

- Potentials of different compounds at standard conditions (1 M, 25°C);
- Unit : Volt (V); 1 J = 1 V × 1 C
- The ability of a compound to reduce is express by a reduction potential measured relative to the reduction potential of hydrogenv (Standard hydrogen electrode, SHE), which is assigned as zero $E^o = 0$ V;
- The lower E^o (more negative) the stronger reducing agent; due to the higher the tendency of the species to donate electrons (be oxidized);

Method II: Using Table of Standard Reduction Potentials (TSRP)

For example, look at the following

$Ag^+ + e \rightleftarrows Ag(s)$ $E^o = +0.80$ V

$Ni^{2+} + 2e \rightleftarrows Ni(s)$ $E^o = -0.25$ V; $Ni(s) \rightleftarrows Ni^{2+} + 2e$ $E^o = +0.25$ V;

When Ag^+ is the oxidant and Ni is the reductant,

$\Delta E = E^o(\text{oxidant}, +) - E^o(\text{reductant}, -) = 0.80\text{ V} - (-0.25\text{ V}) = 1.05$ V, when $\Delta E > 0$ the reaction will be spontaneous; Otherwise, the reaction is non-spontaneous when $\Delta E<0$.

$Cu + AgNO_3 \longrightarrow$ $\Delta E = E^o_{(Cu/Cu2+)} + E^o_{(Ag+/Ag)} = -0.34\text{ V} + 0.80\text{ V} = 0.46$ V

$Cu + Zn(NO_3)_2 \longrightarrow$ $\Delta E = E^o_{(Cu/Cu2+)} + E^o_{(Zn2+/Zn)} = -0.34\text{ V} + (-0.76\text{ V}) = -1.1$ V

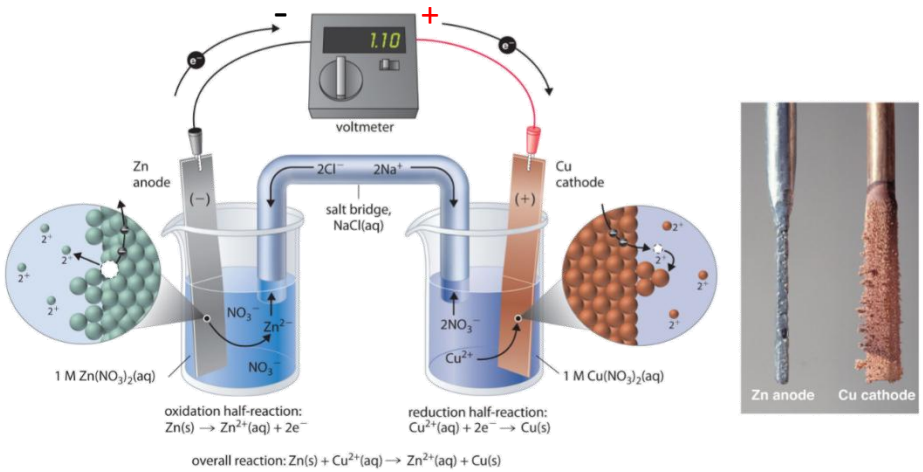
$Fe + Cu(NO_3)_2 \longrightarrow$ $\Delta E = E^o_{(Fe/Fe2+)} + E^o_{(Cu2+/Cu)} = 0.44\text{ V} + 0.34\text{ V} = 0.78$ V

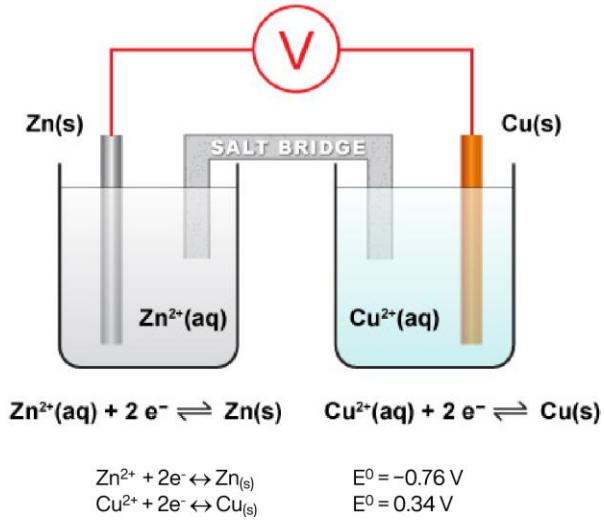
$Al + Cu(NO_3)_2 \longrightarrow$ $\Delta E = E^o_{(Al/Al3+)} + E^o_{(Cu2+/Cu)} = 1.66\text{ V} + 0.34\text{ V} = 2$ V

$Hg + Cu(NO_3)_2 \longrightarrow$ $\Delta E = E^o_{(Hg/Hg2+)} + E^o_{(Cu2+/Cu)} = -0.79\text{ V} + 0.34\text{ V} = -0.45$ V

5. Potentials of galvanic cells

Galvanic cells are cells in which spontaneous electrochemical reactions occur and a chemical energy is converted into electrical energy. Their major use is for batteries and fuel cells.





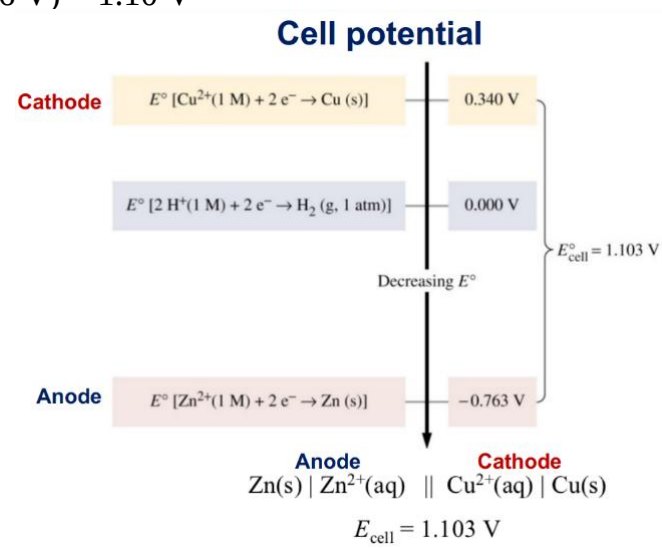
Cell notation: $Zn(s) \mid Zn(NO_3)_2$ (1M, aq) || $Cu(NO_3)_2$ (1M, aq) | Cu(s)

Anode Cathode

" | " — phase interface; " || " — salt bridge.

$E_{Cell} = E^o$(Cathode half-cell, +) – E^o(Anode half-cell, -)

E^o_{Cell} = 0.34 V – (-0.76 V) = 1.10 V



6. Nernst equation for the potential at nonstandard conditions

This equation describes the relationship between the potential and concentration of a compound in solution.

$$a\,A + b\,B \rightleftharpoons c\,C + d\,D \qquad Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

$$E = E^{o} - \frac{0.059}{n}\log Q$$

The potential E depends on the concentrations of the products and reactants and on *n* – The number of electrons participating in the reaction.

6. Electrolytic cells (requires energy) V.S. Galvanic cell (generates energy);

$$Q = I \times t = n_e \times F$$

$$n_e = \frac{I \times t}{F}$$

F is Faraday's constant (96,485 C/mol);

Amount of charge of 1 mol e,

$$1\ e = 1.602 \times 10^{-19}\ C;\ 1\ mol\ e = 6.022 \times 10^{23} \times 1.602 \times 10^{-19}\ C = 9.647 \times 10^{4}\ C$$

n_e is the number of electrons involved in the reaction;

t is operation time (s),

I is the current (A), 1Amper= 1C/s

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Exercise:

T13-1. Balance the following redox chemical equations by first writing each half cell reaction:

a. $Au^{3+}_{(aq)} + I^{-}_{(aq)} \rightarrow Au_{(s)} + I_{2\,(s)}$

b. $Cr_2O^{2-}_{7(aq)} + HNO_{2\,(aq)} \rightarrow Cr^{3+}_{(aq)} + NO^{-}_{3\,(aq)}$ Under acidic conditions

c. $Cr(OH)_{3\,(s)} + ClO^{-}_{3\,(aq)} \rightarrow CrO^{2-}_{4(aq)} + Cl^{-}_{(aq)}$ Under basic conditions

T13-2. Five carbon containing molecules are given:

$CH_2O_{2\,(l)}$; $CH_2O_{\,(g)}$; $CH_4O_{\,(l)}$; $CO_{2\,(g)}$; $CH_{4\,(g)}$

a. Determine the oxidation state (O.S) of the carbon atom in each compound;

b. Consider a chemical reaction where $CH_4O_{\,(l)}$ and $Cr_2O_7^{2-}{}_{(aq)}$ and form $CH_2O_{2\,(l)}$. Which one of these two ions will be an additional product of the reaction: $CrO_4^{2-}{}_{(aq)}$ or $Cr^{3+}{}_{(aq)}$?

c. How many moles of electrons are transferred in the reaction when 1 mol of $CH_2O_{2\,(l)}$ are formed?

d. Oxidation of $CH_4O_{\,(l)}$ is done in two steps. First compound A is formed, and then $CH_2O_{2\,(l)}$. A is one of the given compounds above. Find what is A.

T13-3. Use the Table of Standard Reduction Potentials (TSRP) to answer the following questions:

a. Calculate E° (standard reduction potential) for the following reaction and determine whether it is spontaneous or not:

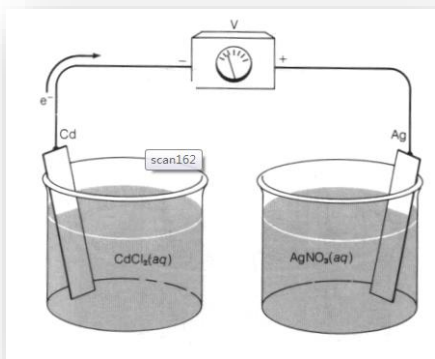
$$2Cr^{3+}_{(aq)} + 2Br^{-}_{(aq)} \rightarrow 2Cr^{2+}_{(aq)} + Br_{2\,(l)}$$

b. Calculate E° (standard reduction potential) for the following reaction and determine whether it is spontaneous or not:

$$3Ag_{(s)} + NO^{-}_{3\,(aq)} + 4H^{+}_{(aq)} \rightarrow NO_{(g)} + 2H_2O_{(l)} + 3Ag^{+}_{(aq)}$$

c. Most disinfectants are killing bacteria by oxidizing them. Considering the standard reduction potentials, which substance is a stronger oxidant, iodine or chlorine?

T13-4. Write the cell notation for the following galvanic cell, and calculate the cell potential in the following if the right half-cell contains 0.50 M $AgNO_3$ (aq) and the left half-cell contains 0.010 M $Cd(NO_3)_2$ (aq).



$E^{o}(Ag^{+}|Ag) = 0.799$ V; $E^{o}(Cd^{2+}|Cd) = -0.402$ V